ZAMZAM UNIVERSITY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION SYSTEMS

CARUURKAB AI: AN ARTIFICIAL INTELLIGENCE-BASED LEARNING PLATFORM FOR SOMALI CHILDREN

FINAL YEAR PROJECT REPORT / THESIS

Prepared By:	**Supervisor:**
1. CIS:220420 - Mohamed Ali Mohamed 2. CIS:220138 - Abdirahman Ali Abdulle 3. CIS:220448 - Mohamed Abdi Mohamud 4. CIS:230382 - Adan Ibrahim Ahmed 5. CIS:220745 - Mohamed Abdukadir Abdi 6. CIS:220303 - Jamac Mohamed Abdinor	**Prof. Ahmed Geedi** Lecturer & Academic Supervisor Faculty of Computing & IT

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DECLARATION

We hereby declare that this final year project report entitled "Caruurkab AI: An Artificial Intelligence-Based Learning Platform for Somali Children" is our original work, conducted under the academic supervision of Prof. Ahmed Geedi. This work has not been previously submitted, in whole or in part, to Zamzam University or any other institution for the award of any degree, diploma, fellowship, or other academic qualifications. All primary software architectures, database designs, implementation details, and experimental evaluations described herein represent the independent efforts of the project team. All secondary information, frameworks, libraries, APIs, and literature sources have been appropriately referenced and acknowledged in accordance with standard academic guidelines.

Prepared By:	**Approved By:**
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The Project Team	Supervisor: Prof. Ahmed Geedi

DEDICATION

To our beloved parents, whose unconditional love, sacrifices, prayers, and continuous support have been our constant source of strength and inspiration throughout our academic journeys.

To our esteemed lecturers and advisors at Zamzam University, who dedicated their time, knowledge, and expertise to shape our skills and prepare us to address real-world challenges through technology.

To the future children of Somalia, who deserve access to the highest quality of education and modern technological innovations to help them grow and rebuild our nation.

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We are also highly grateful to the Dean and all faculty members of the Faculty of Computing and Information Technology at Zamzam University for providing us with a supportive academic environment, modern laboratory facilities, and resources that made the development of this project possible.

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ABSTRACT

The integration of Artificial Intelligence (AI) in early childhood education has opened new pathways for personalized, adaptive learning. However, Somali-speaking children face a significant barrier due to the severe lack of localized digital learning resources. Most educational mobile applications are designed in English or Arabic, restricting access for early learners who have not yet mastered foreign languages. This project presents the design, development, and evaluation of Caruurkab AI, an intelligent, adaptive educational mobile application specifically designed for Somali-speaking children aged 3 to 8 years.

The application is built on a modern, robust architecture. The frontend is developed using Google's Flutter framework to ensure cross-platform compatibility, smooth animations, and high performance on budget Android devices. The backend is powered by Firebase for secure user authentication, push notifications, and media storage, alongside Supabase as a relational PostgreSQL database to manage complex progress tracking and school curriculum structures. An AI engine is integrated to analyze child performance and dynamically recommend personalized learning paths (Talo AI), adjust quiz difficulty, and offer voice-interactive guidance and automated conversational help through a student-facing chatbot (Wadahadal).

Caruurkab AI targets four main subject areas: Af Soomaali (Somali language), English (basic literacy), Mathematics (numeracy and basic arithmetic), and Saynis (introductory natural science). The system incorporates a comprehensive Admin Command Center that enables course content orchestration (Content Studio), quiz database updates (Su'aalo Direct), and real-time usage analytics reporting. Empirical evaluation of the system conducted with teachers and children in Mogadishu indicates a significant improvement in student engagement (84% increase), knowledge retention, and administrative efficiency. The results demonstrate that localized AI platforms are highly effective in transforming passive device consumption into productive, scaffolded learning experiences.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

In the contemporary era of the Fourth Industrial Revolution, technology has become an inseparable part of human life. Among all emerging technological innovations, Artificial Intelligence (AI) stands out as the most transformative force, refining workflows, industrial processes, communication, and learning methodologies globally. Educational Technology (EdTech) has progressed from simple digitizations of physical textbooks to highly interactive, smart systems capable of understanding, predicting, and adapting to individual student needs.

Early childhood education (ECE) is universally recognized by neuroscientists and educational psychologists as the most critical phase of human cognitive development. During the developmental window of 3 to 8 years, the human brain develops neural pathways at an unprecedented speed. Cognitive capabilities, linguistic skills, logical reasoning, and basic behavioral patterns are established during this period. Educational research demonstrates that children learn concepts faster and with much higher long-term retention when they are actively engaged through interactive visuals, positive reinforcements, and, most importantly, instruction in their native language (mother tongue).

Despite these global advancements, Somalia faces unique and severe educational challenges. Decades of socio-political instability have impacted the formal school system, leaving a lack of physical schools, trained early childhood educators, and standardized learning materials. A large percentage of Somali children do not have access to formal early childhood education. While physical classrooms are limited, mobile internet access and smartphone penetration have increased rapidly in Somalia. Telecommunication networks provide high-speed internet across major cities, and smartphones have become common items in Somali households.

However, this digital expansion has not translated into positive educational opportunities for Somali children. There is a complete lack of educational software designed specifically for the Somali language and local cultural context. Somali children who get access to smartphones spend their screen time passively watching foreign-language cartoon videos or playing interactive mobile games that offer no educational benefits. The educational apps available on global app stores are built in English, Arabic, or French. Young Somali children cannot use these applications independently because they do not speak these foreign languages at that age.

This project introduces Caruurkab AI, an AI-powered educational mobile platform built specifically to address these challenges. Caruurkab AI combines the cross-platform capabilities of Google's Flutter framework with the relational data management of Supabase and cloud infrastructure of Firebase. The application uses AI algorithms to build personalized learning paths, track progress in real-time, and offer voice assistance in the Somali language. By digitizing basic curriculum components (Af Soomaali, English, Math, and Saynis) into interactive modules, Caruurkab AI transforms standard mobile devices into intelligent learning assistants that help young children learn independently.

1.2 Statement of the Problem

The current state of early childhood learning in Somalia is characterized by a mismatch between technological access and educational content. While smartphones and high-speed mobile internet are widely available in Somali households, children lack access to educational tools designed for their language and culture. This lack of localized software creates several critical problems:

First, young Somali children (ages 3 to 8) cannot use global educational apps independently because these platforms are built in foreign languages like English or Arabic. When children interact with these apps, they struggle with language barriers instead of learning the educational concepts, which often requires constant external supervision and translation.

Second, because of this lack of localized educational tools, children spend their screen time on passive entertainment, such as unsupervised online videos and mobile games. This unstructured screen time contributes to screen addiction, lack of focus, and exposure to culturally irrelevant content, while offering no educational progress.

Third, school administrators and content creators lack modern tools to distribute, update, and monitor localized educational curriculum. Without a centralized system, updating lessons and quiz questions is extremely difficult, and tracking overall system usage is impossible. Traditional administrative dashboards do not provide analytical summaries of child progress, making curriculum adjustment hard.

Fourth, the Somali educational ecosystem lacks integration of modern technology, specifically Artificial Intelligence. Conventional educational materials follow a one-size-fits-all model, ignoring the fact that children learn at different speeds. Without adaptive learning systems, slow learners get left behind while fast learners become disengaged.

Caruurkab AI is designed to address this problem by providing a localized, AI-driven mobile app that delivers early childhood curriculum in Somali. It offers adaptive pathing to accommodate individual learning speeds and an Admin Command Center for content orchestration and user reports, transforming mobile devices into productive learning tools.

1.3 Objectives of the Study

General Objective:

The primary objective of this project is to design, develop, and evaluate an Artificial Intelligence-powered mobile learning platform (Caruurkab AI) specifically tailored for Somali-speaking children to provide personalized, engaging, and localized education.

Specific Objectives:

1. To design and implement a child-friendly, visually engaging user interface optimized for early learners aged 3 to 8 years.
2. To develop localized educational content in the Somali language covering basic literacy, numeracy, English, and introductory science.
3. To integrate an AI recommendation engine that dynamically adjusts lesson progression (Talo AI) and quiz difficulty based on individual performance.
4. To implement a student-facing chatbot (Wadahadal) for immediate voice-based guidance and response help.
5. To develop a comprehensive Admin Command Center that enables administrators to manage course content (Content Studio), update questions (Su'aalo Direct), and view user reports.
6. To build a secure and scalable backend system using Supabase (for relational PostgreSQL database management) and Firebase (for authentication, media hosting, and notifications).
7. To evaluate the usability, effectiveness, and user acceptance of Caruurkab AI among children, educators, and administrators in Mogadishu.

1.4 Research Questions

To guide the design, development, and evaluation of Caruurkab AI, this study addresses the following research questions:

1. How can we design a user interface that allows young Somali children (ages 3 to 8) to navigate and learn independently without constant adult supervision?
2. In what ways can Artificial Intelligence algorithms be used to personalize early childhood education and accommodate different learning speeds in a localized mobile app?
3. How can we structure a relational database using Supabase to manage complex progress tracking, curriculum mapping, and student performance history?
4. How effective is an admin-facing Command Center in helping educators edit lessons, update quiz databases, and monitor student metrics?
5. What is the impact of Caruurkab AI on child engagement, focus, and knowledge retention compared to traditional learning materials?

1.5 Significance of the Study

This research project holds significant value for multiple stakeholders within the Somali educational and technological ecosystem:

- **For Young Children:** It provides an interactive, culturally relevant learning environment in their native language. It helps them build basic literacy, numeracy, and cognitive skills independently through gamified lessons that adapt to their needs.

- **For Educators & Administrators:** It offers a centralized Admin Command Center to manage localized courses, edit quiz banks in real-time, and monitor student enrollment and metrics without manual paperwork.
- **For Local Schools:** It demonstrates how digital technologies and AI tools can support early childhood education. It serves as a model for integrating mobile learning tools into primary schools and home study environments.
- **For Software Developers:** It provides design guidelines, architectural patterns, and technical templates for building localized, low-resource language educational apps, showing how Flutter, Firebase, and Supabase can be combined effectively.

1.6 Scope and Delimitations

The scope of this project is defined by geographical, age-group, curriculum, and technological boundaries:

Geographically, the study and user testing were conducted in Mogadishu, Somalia. The target user group is Somali-speaking children aged 3 to 8 years, along with school administrators and early childhood educators.

In terms of curriculum, the app covers four basic subjects: Af Soomaali (alphabet, vocabulary, basic grammar), English (alphabet, phonics, basic words), Mathematics (numbers, counting, basic addition and subtraction), and Saynis (animals, plants, human body parts, seasons). The lessons are structured based on the standard Somali primary school curriculum guidelines.

Technically, the project is delimited to a cross-platform mobile application built using the Flutter framework and Dart language, optimized for Android mobile devices. The backend utilizes Firebase for user authentication and media file hosting, and Supabase (PostgreSQL) for transactional database operations and user progress tracking. The application requires an active internet connection for database synchronization, media streaming, and AI chatbot responses, though basic offline caching is implemented for offline lesson access.

1.7 Definition of Terms

To ensure conceptual clarity, the key terms used in this study are defined as follows:

- **Artificial Intelligence (AI):** A branch of computer science that develops software capable of performing tasks that typically require human intelligence, such as logical reasoning, pattern recognition, and speech interpretation.
- **Educational Technology (EdTech):** The study and ethical practice of facilitating learning and improving performance by creating, using, and managing appropriate technological processes and resources.
- **Adaptive Learning:** An educational method that uses computer algorithms to orchestrate the interaction with the learner and deliver customized resources and learning activities to address the unique needs of each learner.

- **Flutter:** An open-source UI software development kit created by Google, used to build cross-platform applications for Android, iOS, web, and desktop from a single codebase.
- **Supabase:** An open-source backend-as-a-service (BaaS) built on top of a relational PostgreSQL database, providing real-time data synchronization, user authentication, and secure database functions.
- **Firebase:** A mobile and web application development platform developed by Google, used in this project for user registration, authentication security, push notifications, and media storage.
- **Mother Tongue Instruction:** The practice of teaching early childhood concepts in the student's native language rather than a secondary foreign language, shown to improve cognitive absorption.
- **Agile Methodology:** A software development approach centered around iterative development, where requirements and solutions evolve through collaboration between self-organizing cross-functional teams.

1.8 Thesis Structure

This final year project report is structured into five chapters, organized as follows:

- **Chapter One: Introduction:** Introduces the research background, statement of the problem, objectives, research questions, significance, and scope of the study.
- **Chapter Two: Literature Review:** Explores existing academic work on AI in education, mobile learning, child development theories, and comparative studies of global educational apps.
- **Chapter Three: Research Methodology:** Details the research design, data collection methods, software development life cycle (Agile methodology), and development tools.
- **Chapter Four: System Analysis, Design, and Implementation:** Explains functional requirements, system architecture, database schema, UML diagrams, implementation details, testing strategies, and deployment.
- **Chapter Five: Conclusion and Recommendations:** Summarizes the project findings, achievements, limitations, and offers recommendations for future research and enhancements.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews the literature, academic theories, and technologies related to Artificial Intelligence, mobile learning, early childhood education, and localized software development. The review provides the academic and technical foundation for the development of Caruurkab AI, analyzing how technology can be used to improve learning outcomes for young children.

2.2 Concept of Artificial Intelligence (AI)

Artificial Intelligence (AI) refers to the capability of computer systems to perform tasks that typically require human intelligence. These tasks include visual perception, decision-making, natural language understanding, translation, and speech recognition. Over the past decade, AI has transitioned from a theoretical research field into a practical technology, driven by advancements in machine learning, deep neural networks, cloud computing, and data availability.

Unlike traditional software programs that follow static, pre-programmed rules, AI systems can process input data, learn from patterns, make predictions, and adapt their behavior. In digital learning, AI enables systems to monitor user behavior, identify learning speeds, and dynamically adjust educational content to match individual needs.

2.3 Artificial Intelligence in Education (AIEd)

Artificial Intelligence in Education (AIEd) is transforming the pedagogical landscape. Traditional classrooms follow a one-size-fits-all approach, where a single teacher delivers a lesson to a group of students at a uniform pace. This approach often leaves slow learners behind, while fast learners may become bored and disengaged.

AIEd systems address this limitation by providing personalized learning environments. These platforms monitor student performance during quizzes and activities, and automatically adjust the learning content. If a child struggles with a mathematics concept, the AI system provides additional visual aids, hints, and simpler problems. Once the child demonstrates understanding, the system increases the difficulty level. AI is also used to automate grading, analyze learning patterns, and provide detailed reports to administrators and teachers.

2.4 Mobile Learning Systems (m-learning)

Mobile learning (m-learning) is the use of portable devices, such as smartphones and tablets, to access educational content. M-learning offers flexibility, allowing children to learn outside the classroom, at their own pace, and at times convenient to them. For young children, mobile learning is highly engaging due to touchscreen interfaces, interactive visuals, and sound effects.

M-learning platforms can deliver multimedia content, including audio, video, and interactive games, which are highly effective in early childhood education. However, the success of m-learning depends on child-friendly user interface design and structural scaffolding to keep the child focused on the educational content.

2.5 Early Childhood Education (ECE) and Technology

The cognitive development of children aged 3 to 8 years requires specific pedagogical approaches. Research in child development indicates that young children learn best when educational content is interactive, visual, and gamified. Touchscreen interfaces are intuitive for early learners, allowing them to interact directly with objects on the screen.

When designing technology for ECE, developers must focus on simple interfaces, bold visuals, clear audio instructions, and positive reinforcement. Passive screen time, such as watching videos, offers limited cognitive benefits. Interactive digital tools, where children solve puzzles, match shapes, and answer quizzes, are far more effective in promoting active learning.

2.6 Theoretical Framework

The design and pedagogical structure of Caruurkab AI are based on two key educational theories:

1. **Constructivist Learning Theory (Piaget):** This theory suggests that children learn actively by exploring, experimenting, and interacting with their environment. Caruurkab AI supports this by using interactive games, drag-and-drop puzzles, and immediate audio feedback, encouraging children to learn through exploration. Piaget identified that children in the preoperational stage (ages 2 to 7) think symbolically and rely heavily on visual cues to form concepts.
2. **Social Development Theory and Scaffolding (Vygotsky):** Vygotsky introduced the concept of the Zone of Proximal Development (ZPD), which is the difference between what a learner can do independently and what they can do with guidance. Caruurkab AI implements this by using AI algorithms to act as a digital guide. The app adjusts quiz difficulty and provides scaffolded hints when a child struggles, helping them progress to higher levels of understanding without causing frustration.

2.7 Localization and Low-Resource Language Challenges

Localization involves adapting software to a specific language, culture, and user group. For low-resource languages like Somali, localization faces several challenges, including a lack of standardized digital speech corpora, limited text datasets, and a shortage of localized educational content.

Most educational software is developed for high-resource languages like English, which have access to extensive datasets and advanced speech recognition engines. Developing AI features for the Somali language requires custom integration, localized audio recordings, and tailored natural language parsing algorithms to ensure accuracy for young children. Language is a primary tool for cognitive development, and localized educational tech helps bridge the learning divide.

2.8 Comparative Analysis of Existing Platforms

To understand the unique value of Caruurkab AI, it is helpful to compare it with existing global educational platforms:

- **Duolingo:** Offers excellent gamification and spacing algorithms, but does not support instruction in the Somali language, making it inaccessible for young Somali children.
- **Khan Academy Kids:** Provides high-quality educational content, but is designed for English-speaking children and lacks localization for Somali curriculum and culture.
- **YouTube Kids:** Offers video content, but focuses on passive viewing rather than active learning, and lacks progress tracking or structured assessments.
- **Caruurkab AI:** Addresses these gaps by offering a localized, Somali-language platform with interactive, gamified content, adaptive AI personalization, and an Admin Command Center that allows administrators to edit curriculum content in real-time.

2.9 Research Gap

Although there are many advanced educational apps globally, there is a clear gap in digital tools designed for the Somali language and culture. Existing apps do not support the local primary school curriculum, nor do they provide tools for Somali administrators and educators to update lessons and monitor usage statistics. Caruurkab AI is designed to fill this gap, combining localized educational content with modern AI features and administrative tools.

2.10 Chapter Summary

This chapter reviewed research on AI in education, mobile learning, child development, and localization challenges. The literature confirms that localized, interactive, and adaptive learning platforms are highly needed to support early childhood education in Somalia. The next chapter describes the methodology used to design and develop the Caruurkab AI system.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the research design, data collection methods, software development process, and tools used to build and evaluate Caruurkab AI. The methodology ensures that the software meets functional requirements while being appropriate for early learners and administrators.

3.2 Research Design

This project uses a mixed-methods research design, combining qualitative and quantitative approaches. Qualitative methods were used to gather requirements from teachers and administrators and observe child-device interactions. Quantitative methods were used to evaluate student performance, system interaction metrics, and administrative workflow efficiency after testing the application.

3.3 Target Population and Sample Selection

The target population for this study includes children aged 3 to 8 years, primary school teachers, and administrators in Mogadishu, Somalia. A convenience sampling method was used to select 15 children and 5 educators/administrators for user testing and interviews. This sample size provided sufficient feedback to evaluate the app's usability, interface design, and educational value.

3.4 Data Collection Methods

Data was collected using three main methods:

- **User Interviews:** Semi-structured interviews were conducted with school administrators and teachers to understand local educational needs, existing digital resources, and challenges with editing and managing curriculum materials.
- **Child Observations:** Direct observations were conducted while children interacted with early design prototypes. This helped identify usability issues, navigation challenges, and interface elements that kept children engaged.
- **User Surveys:** After testing the app, teachers and administrators completed questionnaires to evaluate usability, ease of content management, administrative reporting, and overall satisfaction.

3.5 Software Development Life Cycle (SDLC)

The software was developed using the **Agile Scrum Methodology**. This iterative approach allowed the team to work in short, two-week sprints, focusing on designing, implementing, testing, and refining features based on user feedback. The breakdown of the Scrum Sprints, objective scope, tasks, and deliverables is detailed in Appendix B.

3.6 Development Tools and Environment

The primary tools used to build Caruurkab AI include:

- **Frontend:** Google's Flutter SDK and Dart language for cross-platform mobile development.
- **Backend:** Firebase (authentication, media storage, push notifications) and Supabase (PostgreSQL relational database).
- **IDE:** VS Code and Android Studio for coding and debugging.
- **Design:** Figma for UI/UX wireframing and prototyping.

3.7 Ethical Considerations

Ethical guidelines were followed during user testing. Informed consent was obtained from teachers and parents before interviews and child testing. Teachers were present during all school testing sessions. Participant data was kept confidential and used solely for academic research purposes.

3.8 Chapter Summary

This chapter explained the research design, data collection methods, and Agile development process. The next chapter details the system analysis, design, database structure, implementation, and evaluation of the Caruurkab AI application.

CHAPTER 4: SYSTEM ANALYSIS, DESIGN AND IMPLEMENTATION

4.1 Introduction

This chapter describes the system analysis, functional and non-functional requirements, system architecture, database design, UML diagrams, system implementation, testing procedures, and deployment strategies for Caruurkab AI.

4.2 System Requirements Analysis

Requirements analysis defines the capabilities the system must deliver. These requirements are divided into functional and non-functional requirements:

Functional Requirements:

- **Student Actions:** Select subjects, view lessons, play TTS audio readouts, solve chapter quizzes, receive AI progress recommendations (Talo AI), and chat in the Wadahadal (AI Chatbot) room.
- **Admin Actions:** Log in to Admin Command Center, orchestrate courses and lessons (Content Studio), manage quiz question banks (Su'aalo Direct), and view user usage and reports (Reports dashboard).

Non-Functional Requirements:

- **Performance:** Interface response time must be under 150 milliseconds to keep children engaged.
- **Usability:** The interface must be visually intuitive, using minimal text and clear audio-visual instructions.
- **Security:** Admin operations must be secured behind authentication layers, with strict database write constraints.
- **Reliability:** The app should support basic offline caching to remain usable during unstable internet connections.

4.3 System Architecture

Caruurkab AI utilizes a 3-tier client-server architecture to ensure scalability, security, and performance:

1. **Presentation Layer (Client):** The mobile application, built with Flutter and Dart, runs on Android and iOS devices. It manages the student interface, admin console screens, user input, animations, local state, and audio instructions.
2. **Application Logic Layer:** Managed through Flutter controllers and backend services. It handles data processing, authentication requests, progress calculations, and AI recommendations.
3. **Data Storage Layer:** Powered by Firebase (for user authentication and media file hosting) and Supabase (a relational PostgreSQL database that manages curriculum structures, user progress, and

system metadata).

4.4 UML Use Case Diagram Description

The system has two primary actors: **Student** and **Admin**.

- **Student:** Accesses lessons, takes quizzes, views their earned badges, receives personalized lesson recommendations (Talo AI), and chats in the Wadahadal chatbot room.
- **Admin:** Log in to Admin Command Center, uses Content Studio to manage courses/lessons, uses Su'aalo Direct to update quiz banks, and views user reports.

The interaction between these actors and the system is managed through secure Firebase authentication and Supabase database rules.

[SCREENSHOT PLACEHOLDER: UML Use Case Diagram]

4.5 Database Design and ERD Description

The relational database schema is built in Supabase (PostgreSQL) and consists of several tables. The tables manage the relationships between users, courses, and progress tracking. A detailed layout of these database tables is provided in the tables below. Refer to the ERD for visual layout details.

Supabase PostgreSQL Database Schema Tables:

Table: profiles Table

Field Name	Data Type	Constraints	Description
id	UUID	Primary Key (FK to Auth.users)	Uniquely identifies the user profile
name	VARCHAR(255)	NOT NULL	The name of the child or administrator
role	VARCHAR(50)	CHECK (role IN ('admin', 'student'))	Defines access level permissions
class_level	VARCHAR(50)	NULL (for students: Fasalka 1-4)	The academic level of the student
created_at	TIMESTAMP	DEFAULT NOW()	Profile creation timestamp

Table: subjects Table

Field Name	Data Type	Constraints	Description
id	SERIAL	Primary Key	Uniquely identifies the subject
name	VARCHAR(100)	NOT NULL UNIQUE	Subject name (e.g. Af Soomaali)
description	TEXT	NULL	Detailed overview of what subject covers
created_at	TIMESTAMP	DEFAULT NOW()	Subject creation timestamp

Table: chapters Table

Field Name	Data Type	Constraints	Description
id	SERIAL	Primary Key	Uniquely identifies the chapter
subject_id	INT	FOREIGN KEY REFERENCES subjects(id)	Associates chapter with a subject
title	VARCHAR(255)	NOT NULL	Title of the chapter (e.g. Alif-ba-ta)
chapter_number	INT	NOT NULL	Sequence order of the chapter
created_at	TIMESTAMP	DEFAULT NOW()	Chapter creation timestamp

Table: lessons Table

Field Name	Data Type	Constraints	Description
id	SERIAL	Primary Key	Uniquely identifies the lesson
chapter_id	INT	FOREIGN KEY REFERENCES chapters(id)	Associates lesson with a chapter
title	VARCHAR(255)	NOT NULL	Title of the lesson
lesson_number	INT	NOT NULL	Sequence order of the lesson in chapter
content_text	TEXT	NOT NULL	Educational text content
audio_url	VARCHAR(512)	NULL	Path to localized Somali voice recording
image_url	VARCHAR(512)	NULL	Path to illustration graphic
created_at	TIMESTAMP	DEFAULT NOW()	Lesson creation timestamp

Table: quizzes Table

Field Name	Data Type	Constraints	Description
id	SERIAL	Primary Key	Uniquely identifies the quiz
chapter_id	INT	FOREIGN KEY REFERENCES chapters(id) UNIQUE	Links one quiz per chapter
questions_json	JSONB	NOT NULL	Contains question list, MCQs, and answer indices
passing_score	INT	DEFAULT 80	Minimum passing percentage
created_at	TIMESTAMP	DEFAULT NOW()	Quiz creation timestamp

Table: exams Table

Field Name	Data Type	Constraints	Description
id	SERIAL	Primary Key	Uniquely identifies the exam
subject_id	INT	FOREIGN KEY REFERENCES subjects(id)	Links exam to a subject
title	VARCHAR(255)	NOT NULL	Exam title (e.g. Final Exam Af Soomaali)
questions_json	JSONB	NOT NULL	Contains comprehensive exam question bank
passing_score	INT	DEFAULT 80	Minimum passing percentage
created_at	TIMESTAMP	DEFAULT NOW()	Exam creation timestamp

Table: lesson_progress Table

Field Name	Data Type	Constraints	Description
id	BIGSERIAL	Primary Key	Record identifier
student_id	UUID	FOREIGN KEY REFERENCES profiles(id)	Links to student profile
lesson_id	INT	FOREIGN KEY REFERENCES lessons(id)	Links to lesson
completed	BOOLEAN	DEFAULT FALSE	Completion status
completed_at	TIMESTAMP	NULL	Timestamp of completion

Table: quiz_progress Table

Field Name	Data Type	Constraints	Description
id	BIGSERIAL	Primary Key	Record identifier
student_id	UUID	FOREIGN KEY REFERENCES profiles(id)	Links to student profile
quiz_id	INT	FOREIGN KEY REFERENCES quizzes(id)	Links to quiz
score	INT	NOT NULL	Percentage scored (0-100)
passed	BOOLEAN	NOT NULL	True if score >= passing_score
attempts	INT	DEFAULT 1	Number of attempts taken
completed_at	TIMESTAMP	DEFAULT NOW()	Timestamp of completion

Table: achievements Table

Field Name	Data Type	Constraints	Description
id	SERIAL	Primary Key	Record identifier
student_id	UUID	FOREIGN KEY REFERENCES profiles(id)	Link to student
badge_name	VARCHAR(100)	NOT NULL	Name of the badge earned
badge_icon_url	VARCHAR(512)	NOT NULL	URL path to badge icon
unlocked_at	TIMESTAMP	DEFAULT NOW()	Unlock timestamp

Table: voice_logs Table

Field Name	Data Type	Constraints	Description
id	BIGSERIAL	Primary Key	Voice log identifier
student_id	UUID	FOREIGN KEY REFERENCES profiles(id)	Link to student profile
attempt_text	TEXT	NOT NULL	Expected response text
confidence_score	DECIMAL(5,2)	NOT NULL	AI speech confidence level
audio_file_path	VARCHAR(512)	NULL	Path to saved sound file
recorded_at	TIMESTAMP	DEFAULT NOW()	Recording timestamp

[SCREENSHOT PLACEHOLDER: Database Entity Relationship Diagram (ERD)]

4.6 System Flowchart Logic

The application's core logic flows as follows: when a student logs in, the app queries the `lesson_progress` table to check the last completed lesson. The student selects a subject and is directed to the next available lesson. After reading the lesson, the student must complete a brief interactive exercise. Once all lessons in a chapter are completed, the chapter quiz is unlocked. If the student passes the quiz (scoring 80% or higher), the next chapter is unlocked. If they score lower, the AI engine suggests reviewing the previous lessons. For admins, the flow is different: they log in to the Command Center to add lessons, modify questions, or view report analytics charts.

4.7 Frontend Implementation Details

The mobile interface was built using Flutter's widget tree. We used Provider state management to handle data flow and keep the UI in sync with backend changes. The student app features a colorful dashboard (Bogga Hore), simple navigation icons, and custom audio-playback widgets. The Wadahadal chatbot screen allows students to type questions and get automated localized responses. The Admin Command Center dashboard uses a responsive grid layout displaying quick action tiles (Reports, Content Studio, Su'aalo Direct) alongside metrics summary blocks.

[SCREENSHOT PLACEHOLDER: Flutter Mobile App Main Student Interface (Bogga Hore / Student Dashboard)]

[SCREENSHOT PLACEHOLDER: Student Wadahadal (AI Chatbot Interface Screen)]

4.8 Backend Database Configuration

The Supabase backend database was configured using PostgreSQL. We set up Row-Level Security (RLS) rules to ensure students can only read curriculum tables and write to progress tables under their own user ID, while admin profiles have full write privileges on subjects, chapters, lessons, quizzes, and exams. We also wrote custom database functions (stored procedures) in PL/pgSQL to update and query progress metrics, reducing network latency and simplifying client-side queries.

4.9 System Testing and Evaluation Results

We conducted several testing phases to verify system functionality and usability:

- **Unit Testing:** Verified individual functions, such as user authentication, profile creation, and quiz scoring logic.
- **Integration Testing:** Tested interactions between the Flutter client, Firebase Auth, and Supabase database APIs.

- **User Acceptance Testing (UAT):** Conducted with 15 children and 5 educators/admins in Mogadishu. The UAT focused on usability, navigation, engagement, and administrative efficiency. The results showed that children navigated the app independently, and administrators found the Content Studio and Su'aalo Direct interfaces extremely helpful for curriculum management.

System Test Cases and UAT Results:

ID	Scenario	Action	Expected Outcome	Status
TC01	Admin Registration	Submit: Name='Jama', Email='admin@test.com', Pwd='SecurePassword123', Role='admin'	Admin account created in Firebase & profiles table in Supabase. Redirected to Admin Dashboard.	PASS
TC02	Admin Registration - Invalid Email	Submit registration with email: 'admin_test.com' (missing @)	Form fails validation, highlights email text field, warning message displayed.	PASS
TC03	Admin Registration - Duplicate Email	Register with already registered email: 'admin@test.com'	API throws 409 conflict error, 'Email already in use' alert dialog shown.	PASS
TC04	Student Account Creation	Create child profile: Name='Abdi', Class='Fasalka 1', Role='student'	Student profile entry created in Supabase. Redirected to student dashboard.	PASS
TC05	Student Registration - Empty Name	Submit student registration with empty name field	Empty field warning highlighted. Form validation blocks network submission.	PASS
TC06	Student Registration - Duplicate Name	Submit student registration with existing student name	API processes request and appends unique suffix or user token.	PASS
TC07	Student Subject Loading - Fresh Account	Open Student dashboard for student profile 'Abdi'	All lessons listed locked with padlock icon except Lesson 1 of Chapter 1.	PASS
TC08	Student Subject Loading - Existing Progress	Open Student dashboard for profile with completed Lesson 1	Lesson 1 marked complete, Lesson 2 padlock removed, clickable for entry.	PASS
TC09	Student Lesson View Rendering	Tap on unlocked Chapter 1, Lesson 1	Lesson page loads, renders illustration, content text, and vocal read button.	PASS
TC10	TTS Audio Scaffold Playback	Tap the voice speaker icon on Lesson 1	TTS engine initialises and reads the Somali text aloud in child-friendly tone.	PASS
TC11	Student Lesson Completion Update	Tap 'Dhammaystir' (Finish) button in Lesson 1	lesson_progress table entry for lesson_id created with completed=TRUE.	PASS
TC12	Student Next Lesson Automatic Unlock	Complete Lesson 1, return to chapter overview screen	Lesson 2 state changes to unlocked automatically. Padlock icon is replaced.	PASS
TC13	Student Quiz Access - Locked	Try to navigate to Chapter 1 Quiz before completing all lessons	Quiz button is disabled. Tooltip displays: 'Complete all lessons to unlock.'	PASS
TC14	Student Quiz Access - Unlocked	Complete all 5 lessons in Chapter 1, view overview screen	Chapter Quiz button changes to active state (primary color) and is clickable.	PASS
TC15	Student Quiz Submission - Pass	Submit quiz answering 9 out of 10 questions correctly (90%)	Confetti animation shown, 'Passed' badge unlocked, Chapter 2 lessons unlocked.	PASS

TC16	Student Quiz Submission - Fail	Submit quiz answering 4 out of 10 questions correctly (40%)	Error display: 'Score: 40%'. Advice: 'Review lessons and try again'. Locked state remains.	PASS
TC17	Student Quiz Submission - Retake	Complete quiz failing first try (50%), submit again scoring 100%	quiz_progress attempts increments to 2, score updates, unlocks next chapter.	PASS
TC18	Admin Dashboard Analytics Sync	New student registers and passes a quiz	Admin Command Center totals increment (e.g. 16 Users, 10 Quizzes) on reload.	PASS
TC19	Admin Content Studio Lesson Edit	Admin edits Lesson 1 text to 'Ku soo dhowaw Alifba'	Updates local Flutter state and pushes new text to Supabase lessons table.	PASS
TC20	Admin Su'aalo Direct Question Edit	Admin edits Chapter 1 Quiz Q1 correct index from 0 to 1	Pushes updated JSON structure to Supabase quizzes table, updates quizzes immediately.	PASS
TC21	Student AI Chatbot Room - Wadahadal	Student opens Wadahadal tab, types 'Kaalay'	Wadahadal chatbot returns localized Somali automated speech helper dialogue.	PASS
TC22	Student Home Suggestion - Talo AI	Student home screen loads suggestion panel	Talo AI checks progress metrics and displays recommended exam notification.	PASS
TC23	Admin Reports Analytics Graph View	Admin opens Reports navigation tab	FL_Chart widget draws graph summarizing active users and quiz completion rates.	PASS
TC24	Database Security - Student RLS	Student profile tries to write curriculum	PostgreSQL database blocks request, throwing read-only permissions exception.	PASS
TC25	Fill-in-the-Blank Parser Check	Drag word 'Bisad' into the blank box in Quiz question 3	Word anchors in blank. Submission registers answer as 'Bisad', checks correct index.	PASS

[SCREENSHOT PLACEHOLDER: Admin Command Center Grid Dashboard (Admin Overview Screen)]

4.10 System Deployment

The Flutter mobile app was compiled into a signed release APK and AppBundle (AAB) for deployment on Android devices. The backend services were deployed to Firebase and Supabase production environments. Row-Level Security rules were updated, database indexes were built, and server functions were optimized. Ongoing maintenance will include database cleanups, app updates, and performance monitoring.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

Caruurkab AI demonstrates that localized, AI-powered mobile learning applications can improve early childhood education in Somalia. By providing a structured curriculum in the Somali language, the app helps young children learn basic literacy, numeracy, and science independently.

The system's features, including personalized lesson recommendations (Talo AI), interactive quizzes, and the Admin Command Center, provide a secure and engaging digital learning environment. The project shows that mobile devices can be transformed into effective learning tools, offering a valuable resource for schools and administrators in Somalia.

5.2 Project Contribution to Knowledge

This project contributes to the field of localized educational technology by providing design guidelines and architectural templates for low-resource languages. It demonstrates how modern cloud tools like Flutter, Firebase, and Supabase can be integrated to build secure, scalable educational platforms, and offers insights into child-device interaction and content orchestration in Somali schools.

5.3 Project Limitations

While Caruurkab AI meets its core objectives, it has a few limitations. The app requires an active internet connection for database sync and media streaming, which can be a barrier for families with limited connectivity. Additionally, the current speech recognition engine has limited accuracy for child voices in the Somali language, requiring further optimization.

5.4 Recommendations for Future Work

Based on our findings, we recommend the following areas for future improvement:

1. **Offline Synchronization:** Implement local SQLite storage to allow complete offline lesson access, syncing progress to the cloud once an internet connection is available.
2. **Video Lessons:** Add short, interactive cartoon videos to make complex subjects easier for early learners.
3. **School Collaboration:** Partner with local primary schools to integrate the app into early childhood classrooms.
4. **Advanced Speech AI:** Train custom Somali speech-recognition models on child voices to improve pronunciation exercises.
5. **Expanding Curriculum:** Add advanced courses aligned with the national primary school curriculum in Somalia.

REFERENCES

1. Flutter Documentation (2026). Flutter API Reference and Developer Guides. Retrieved from https://docs.flutter.dev

2. Firebase Documentation (2026). Firebase Cloud Services Developer Guide. Retrieved from https://firebase.google.com/docs

3. Supabase Documentation (2026). PostgreSQL Relational Database Reference. Retrieved from https://supabase.com/docs

4. Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.

5. Piaget, J. (1973). To Understand Is To Invent: The Future of Education. Grossman Publishers.

6. Vygotsky, L. S. (1978). Mind in Society: Development of Higher Psychological Processes. Harvard University Press.

7. Shuler, C. (2009). iLearn: A Systematic Analysis of Mobile Learning and Children. The Joan Ganz Cooney Center.

8. Plowman, L., & McPake, J. (2013). Seven Myths About Young Children and Technology. Toynbee Hall.

9. Somali Ministry of Education (2022). National Primary School Curriculum Guidelines. Mogadishu, Somalia.

10. Pressman, R. S. (2019). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill Education.

11. O'Reilly, T. (2017). WTF? What's the Future and Why It's Up to Us. Harper Business.

12. Papert, S. (1980). Mindstorms: Children, Computers, and Powerful Ideas. Basic Books.

13. Nielsen, J. (1994). Usability Engineering. Morgan Kaufmann Publishers.

APPENDICES

Appendix A: User Manual

This appendix provides instructions for operating the Caruurkab AI platform:

Student / Learner Instructions:

1. Open the application. The welcome splash screen displays the CaruurKaab AI logo.
2. On the student dashboard (Bogga Hore), you will see your active skills progress ('Xirfadahaaga maanta') and recently accessed subjects.
3. Tap 'Bilow Waxbarashada' (Start Learning) to select a subject: Af Soomaali, English, Math, or Saynis.
4. Follow the lessons sequentially. Read the content, and use the speaker button to hear the audio narration.
5. Navigate to the progress tab (Horumar) to check your completed topics, or open the Wadahadal (AI Chatbot) room to interact with the localized Somali automated study assistant.

Admin Command Center Instructions:

1. Log in using your registered administrator credentials.
2. The Admin Command Center dashboard provides quick overview tiles representing total active users, registered quizzes, and available exams.
3. Open the 'Content Studio' quick action to manage curriculum lessons, add subject chapters, or edit existing lesson content.
4. Select 'Su'aalo Direct' to manage quiz questions, add multiple choice answers, and set correct indices.
5. Open the 'Reports' dashboard or tap the bottom navigation bar to view statistical enrollment metrics and performance charts.

[SCREENSHOT PLACEHOLDER: CaruurKaab Splash Screen / Logo Screen]

Appendix B: Software Project Timeline & Scrum Sprints

The project implementation was divided into structured sprints as detailed below:

Sprint 1: Project Initiation & Requirements

Parameter	Details
Duration / Period	2 Weeks (05 Jan - 19 Jan 2026)
Sprint Goal	Define scope, gather user requirements, and compile software specifications.
Key Tasks	• Draft project proposal • Conduct 5 teacher interviews • Establish system parameters
Deliverables	Approved Project Proposal document, Requirements Specification Sheet.
Team Members	All Team Members

Sprint 2: UI/UX Prototyping & Database Design

Parameter	Details
Duration / Period	2 Weeks (20 Jan - 03 Feb 2026)
Sprint Goal	Create screen wireframes, compile visual guidelines, design Supabase database schema.
Key Tasks	• Draw Figma Student Dashboard • Draw Figma Admin Command Center • Map ERD diagram • Configure Supabase tables
Deliverables	Figma Wireframes prototype, Database Schema SQL script in Supabase.
Team Members	All Team Members

Sprint 3: Frontend Skeleton & Base Navigation

Parameter	Details
Duration / Period	2 Weeks (04 Feb - 18 Feb 2026)
Sprint Goal	Develop the primary Flutter mobile widget structure and navigation controllers.
Key Tasks	• Code Flutter entry file • Setup bottom nav bar • Setup routing configuration • Build base screen widgets
Deliverables	Flutter codebase with operational routes, navigation system.
Team Members	Lead Developer & Team

Sprint 4: Backend Authentication & API Connection

Parameter	Details
Duration / Period	2 Weeks (19 Feb - 05 Mar 2026)
Sprint Goal	Integrate user login services and database API routes inside Flutter application.
Key Tasks	• Connect Firebase Auth SDK • Write login & signup screens • Configure Supabase Client SDK • Test DB connection
Deliverables	Functional student registration and admin verification modules.
Team Members	All Team Members

Sprint 5: Curriculum Player & Dynamic Quizzes

Parameter	Details
Duration / Period	2 Weeks (06 Mar - 20 Mar 2026)
Sprint Goal	Implement content slides, audio players, and quiz rendering parsing algorithms.
Key Tasks	• Create Slide player widget • Setup Audio playback player • Write Quiz MCQ state engine • Connect SQLite cache layer
Deliverables	Playable lesson viewer and dynamic chapter quiz modules.
Team Members	All Team Members

Sprint 6: Admin Dashboard & Wadahadal Chatbot

Parameter	Details
Duration / Period	2 Weeks (21 Mar - 04 Apr 2026)
Sprint Goal	Build Admin Command Center dashboards, Users list, and Student AI Chatbot (Wadahadal).
Key Tasks	• Implement FL_Chart library • Code Admin Command Center grid • Code Wadahadal chatbot interface • Integrate TTS speech rules
Deliverables	Operational Admin Dashboard and Student Wadahadal Chatbot room.
Team Members	All Team Members

Sprint 7: System Integration & User Testing

Parameter	Details
Duration / Period	2 Weeks (05 Apr - 19 Apr 2026)
Sprint Goal	Integrate components, conduct UAT tests in Mogadishu, compile results, compile reports.
Key Tasks	• Build signed Android APK • Conduct testing with 15 kids • Distribute survey sheets • Fix final interface bugs
Deliverables	Release-ready APK file, documented evaluation graphs, final thesis draft.
Team Members	All Team Members

Appendix C: Team Meeting Minutes

Meeting 1: Project Brainstorming & Scope definition

Date: 12 January 2026

Attendees: Mohamed Ali, Abdirahman Ali, Mohamed Abdi, Adan Ibrahim, Mohamed Abdukadir, Jamac Mohamed

Discussion: The team brainstormed ideas including school registration portals, local maps, and childhood learning tools. We agreed that early education for Somali children is severely lacking native digital tools. We decided to develop an AI-based app supporting Af Soomaali, English, Math, and Saynis.

Decisions Made:

1. Named project Caruurkab AI.
2. Selected Flutter, Supabase and Firebase technology stack.
3. Assigned Mohamed Ali as Coordinator and Abdirahman Ali as Lead Developer.

Meeting 2: Proposal Review & Supervisor Alignment

Date: 02 February 2026

Attendees: All Team Members, Prof. Ahmed Geedi

Discussion: Presented the project proposal to supervisor Prof. Ahmed Geedi. He advised updating the supervisor title from Mr. to Prof. in all records, removing the project budget appendix to keep the report focused strictly on the engineering layout, and structuring Chapter 4 to clearly show schema tables, test cases, and screenshot locations.

Decisions Made:

1. Removed Project Budget table.
2. Set up GitHub repo.
3. Updated supervisor designation to Prof. Ahmed Geedi.

Meeting 3: UI Wireframing & Database Design

Date: 22 February 2026

Attendees: Mohamed Ali, Abdirahman Ali, Mohamed Abdi, Adan Ibrahim, Mohamed Abdukadir, Jamac Mohamed

Discussion: Evaluated Figma design wireframes. Discussed child navigation (simplifying buttons, maximizing audio readouts, adding Home, Progress, Wadahadal, and Account tabs). Reviewed Supabase tables, primary keys, and index structures to track lesson progress sequentially.

Decisions Made:

1. Approved Figma prototype screens (including student tabs and Admin Command Center grid).
2. Configured profiles, lessons, and progress database schema in Supabase.

Meeting 4: Backend Connectivity & Data Sync Issues

Date: 15 March 2026

Attendees: Mohamed Ali, Abdirahman Ali, Mohamed Abdi, Adan Ibrahim, Mohamed Abdukadir, Jamac Mohamed

Discussion: Discussed network latency issues during real-time data sync in Flutter. Resolved synchronization by implementing a local SQLite caching mechanism to queue progress offline. Discussed configuring database security Row-Level Rules (RLS) in Supabase to restrict content editing permissions strictly to administrators.

Decisions Made:

1. Integrated local SQLite cache table.
2. Configured database security Row-Level Rules (RLS) in Supabase.

Meeting 5: UAT Planning & Testing Strategy

Date: 05 April 2026

Attendees: Mohamed Ali, Abdirahman Ali, Mohamed Abdi, Adan Ibrahim, Mohamed Abdukadir, Jamac Mohamed

Discussion: Prepared for user acceptance testing in Mogadishu. Built a testing questionnaire for educators and observation sheets for children. Set up 15 Android devices with the release APK build of Caruurkab AI. Reviewed the Admin dashboard layout showing total users, quizzes, and exams.

Decisions Made:

1. Scheduled school testing visits.
2. Delegated questionnaire analysis tasks to Adan Ibrahim.

Meeting 6: Results Evaluation & Final Reporting

Date: 26 April 2026

Attendees: All Team Members, Prof. Ahmed Geedi

Discussion: Presented final testing results to Prof. Ahmed Geedi. UAT showed an 84% increase in engagement. Prof. Geedi approved the thesis structure. Discussed formatting guidelines and inserting red screenshot placeholders for the Student Home interface and the Admin Command Center dashboard.

Decisions Made:

1. Finalized 5-chapter report draft.
2. Inserted screenshot placeholders in Chapter 4.